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Distributed observer–controller co–design for string stability in vehicle platoons

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ABSTRACT

This paper presents a co-design framework for distributed observers and controllers to ensure the string stability of vehicle platoons under external disturbances. Traditional distributed controllers often rely on the explicit knowledge of desired spacing to generate control signals. To eliminate this dependency, we design a distributed observer for each vehicle to provide a comprehensive perception of the entire platoon's status. By leveraging locally measurable outputs and inter-vehicle communication, each observer estimates the tracking errors of all vehicles relative to a global desired trajectory. These estimates are then directly utilized for control signal generation, removing the need for local spacing calculations. Furthermore, unlike conventional observers that require access to the private control inputs of all vehicles in the platoon, our design enables vehicles to estimate other vehicles' inputs based on their own estimated states. To ensure synergistic performance, string stability and observer robustness are integrated into a single weighted $\mathcal{H}_\infty$ performance index. The coupling terms between the controller and observer are decoupled using Young's inequality, facilitating a simultaneous design of both components. The proposed method is validated in Quanser Interactive Labs (QLabs) using four 1/10-scale QCar2 vehicles. Validation scenarios include constant-speed cruising on smooth and wavy roads, as well as start-stop maneuvers. The results demonstrate that the proposed approach effectively maintains string stability and prevents collisions, even when faced with external disturbances, such as those caused by uneven terrain and rapid velocity changes.

1. Introduction

A vehicle platoon refers to a group of vehicles that travel closely at a set cruising speed while maintaining desired inter-vehicle spacing (Guo & Yue, 2012). This coordination improves traffic throughput, reduces fuel consumption, and enhances road safety (Dutta et al., 2022; Wang et al., 2023). However, external disturbances, such as wind gusts, road irregularities, and sensor noise, can significantly degrade platoon performance and safety (Bessafa et al., 2026). Consequently, string stability, the ability to prevent the amplification of disturbances along the vehicle platoon, is a critical requirement in platoon control (Rajamani et al., 2000; Zeng et al., 2019). This property is particularly vital when a platoon navigates uneven terrain or changes velocity rapidly. In such scenarios, vehicles often apply aggressive control actions to quickly minimize spacing errors, which can inadvertently lead to oscillations or even collisions if the system is not string-stable. Therefore, ensuring that disturbances do not propagate backward is the cornerstone of reliable and safe platoon management.

To achieve string stability, controllers can be built from locally measurable outputs (e.g., static output feedback in Viadero-Monasterio et al., 2025). Performance can be further improved by vehicle-to-vehicle (V2V) communication, which enables distributed control by sharing information among neighboring vehicles (Coppola et al., 2022; Wang et al., 2024; Zheng et al., 2016). However, a significant number of existing distributed controllers rely explicitly on desired spacing to generate control signals. While this dependency is manageable under a constant spacing (CS) policy, where the desired gaps are fixed, it poses challenges in more dynamic scenarios.

In many practical cases, the host vehicle cannot compute the desired spacing using only local data and neighbor-to-neighbor V2V communication. For instance, under a Constant Time Headway (CTH) policy, the desired gap between a host vehicle and the leader often depends on the cumulative velocities of all intermediate vehicles. Since a vehicle typically only communicates with its neighbors rather than the entire platoon, it may lack the information necessary to calculate a reliable desired spacing locally. Given that the ultimate purpose of spacing

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calculations is to derive tracking errors for the controller, an intuitive alternative is to estimate these errors directly (Huang et al., 2024). This shift in perspective eliminates the need for explicit spacing calculations. Recently, distributed observers have provided a solid theoretical foundation for this approach (Meng et al., 2025a,b; Wang & Morse, 2018; Yang & Dong, 2024). By equipping each vehicle with a distributed observer that leverages local measurements and limited V2V communication, it becomes possible to estimate the states of the entire platoon, thereby facilitating error-based control without predefined spacing constants.

Traditional distributed observers typically require real-time access to the control inputs of all agents within the network to guarantee state convergence. However, in practical platooning scenarios, vehicles are often reluctant to share their specific control signals due to privacy and cybersecurity concerns. To address this, some researchers have treated the control inputs of other agents as unknown variables, employing distributed Unknown Input Observers (UIO) to simultaneously estimate states and reconstruct control inputs (Yang et al., 2022). While promising, this approach often relies on stringent rank conditions that are difficult to satisfy in vehicle platoons. Despite these limitations, the concept of estimating rather than communicating neighbor inputs has inspired the work presented in this paper. Instead of relying on the strict structural assumptions required by conventional UIOs, we propose an alternative mechanism to account for the control inputs of neighboring vehicles.

To overcome the aforementioned limitations, this paper proposes a co-design strategy for distributed observers and controllers. This approach allows the observer to estimate the control inputs of other vehicles based solely on their estimated states, ensuring the convergence of estimation errors without requiring private information. However, this estimation mechanism introduces complex coupling terms between the control gains and the observer's error dynamics. Furthermore, balancing observer robustness against external disturbances with the string stability of the platoon remains a significant challenge. To address this, we formulate a weighted $\mathcal{H}_\infty$ performance index that simultaneously minimizes the impact of disturbances on estimation accuracy and platoon coordination. By employing Young's inequality, the nonlinear coupling terms are decoupled, transforming the co-design problem into a convex optimization framework based on Linear Matrix Inequalities (LMIs).

The main contributions are summarized as follows:

- We develop a simultaneous design for distributed observers and controllers that eliminates the explicit dependency on desired spacing and the need for private control inputs from neighboring vehicles, thereby enhancing both flexibility and privacy.
- A weighted $\mathcal{H}_\infty$ performance is proposed to unify observer robustness and platoon string stability. Through the application of Young's inequality, the interdependent design variables are decoupled, resulting in a set of solvable LMIs that guarantee system-wide stability under external disturbances.
- Beyond numerical simulations, the proposed method is validated in Quanser Interactive Labs (QLabs) using 1/10-scale QCar2 vehicles. The validations cover diverse and challenging conditions, including wavy terrain and frequent start-stop maneuvers, demonstrating the effectiveness of the algorithm.

The remainder of the paper is organized as follows. Section 2 presents the problem formulation and preliminaries. Section 3 details the distributed observer–controller co–design method. Section 4 reports validation results in QLabs at the different scenarios. Finally, Section 5 concludes the paper.

2. Problem formulation and preliminaries

In this section, we introduce some notations and preliminaries used in this paper. Then, the problem formulation is presented to illustrate our motivation.

2.1. Preliminaries and notation

The following notations are used throughout this paper. The symbol $\otimes$ denotes the Kronecker product. $(*)$ indicates block entries determined by symmetry. $S^\top$ denotes the transpose of S. $\mathbb{I}_r$ denotes the $r\times r$ identity matrix. The vector $\mathbf{1}_n \in \mathbb{R}^n$ denotes the n-dimensional all-ones vector. For a square matrix S, $S>0$ ($S<0$) means that S is positive definite (negative definite). For a symmetric positive definite matrix S, $\lambda_{\max}(S)$ and $\lambda_{\min}(S)$ denote the maximum and minimum eigenvalues of S, respectively. $\mathrm{Sym}(S)$ denotes $S+S^\top$. $\underset{N}{\mathrm{diag}}(S)$ denotes $\mathrm{diag}(S,S,\dots,S)$ with N copies of S.

The communication topology of the platoon is modeled as a directed graph. A graph $\mathcal{G}=(\mathcal{V},\mathcal{E})$ consists of the node set $\mathcal{V}=\{1,\dots,N\}$ and the edge set $\mathcal{E}\subset\mathcal{V}\times\mathcal{V}$ (ordered pairs of nodes, i.e., edges). The graph $\mathcal{G}$ is directed when the pairs in $\mathcal{E}$ are ordered, and it is strongly connected if, for any distinct nodes i and j, there exists a directed path from i to j. The adjacency matrix $\mathcal{A}=[a_{ij}]\in\mathbb{R}^{N\times N}$ is defined by $a_{ij}=1$ if $(i,j)\in\mathcal{E}$ and $a_{ij}=0$ otherwise. The Laplacian matrix is $\mathcal{L}=\mathcal{D}-\mathcal{A}$, where $\mathcal{D}=[d_{ij}]\in\mathbb{R}^{N\times N}$ is the degree matrix with $d_{ij}=0$ for $i\neq j$ and $d_{ii}=\sum_{j=1}^N a_{ij}$. The following lemma states an important property of the Laplacian matrix.

Lemma 1. *Olfati-Saber and Murray (2004) If $\mathcal{G}$ is a strongly connected directed graph, then there exists a unique positive row vector $r=\begin{bmatrix} r_1 & r_2 & \cdots & r_N\end{bmatrix}$ satisfying $r\mathcal{L}=0$ and $r\mathbf{1}_N=N$. Defining $R=\mathrm{diag}(r_1,\dots,r_N)$, the symmetric matrix $\hat{\mathcal{L}}=R\mathcal{L}+\mathcal{L}^\top R$ is positive semidefinite. Furthermore, $\mathbf{1}_N^\top\hat{\mathcal{L}}=0$ and $\hat{\mathcal{L}}\mathbf{1}_N=0$, which implies that $\lambda_1=0$ is an eigenvalue of $\hat{\mathcal{L}}$, and the remaining eigenvalues of $\hat{\mathcal{L}}$ are positive real.*

The following lemma, based on Schur's complement lemma and Young's inequality, is useful for the subsequent analysis

Lemma 2. *For given matrices $X\in\mathbb{R}^{m\times n}$ and $Y\in\mathbb{R}^{n\times m}$, and a symmetric negative definite matrix $Z\in\mathbb{R}^{m\times m}$, if there exists a symmetric positive definite matrix $S\in\mathbb{R}^{n\times n}$ such that the following inequality:*

$$\begin{bmatrix} Z & XS & Y^\top \\ * & -\frac{1}{\iota}S & 0 \\ * & * & -\iota S\end{bmatrix}\leq 0, \tag{1}$$

where $\iota>0$, then the following inequality holds:

$$M=Z+\mathrm{Sym}(XY)\leq 0. \tag{2}$$

Proof. Using the Schur lemma on (1), we can get

$$Z+\iota XSX^\top+\frac{1}{\iota}Y^\top S^{-1}Y\leq 0. \tag{3}$$

Applying Young's inequality to (2), we obtain

$$M\leq Z+\iota XSX^\top+\frac{1}{\iota}Y^\top S^{-1}Y\leq 0. \tag{4}$$

This completes the proof. □

This lemma decouples the product term XY by introducing the term XS, which is useful for the subsequent controller and observer design.

2.2. Problem formulation

In this paper, we consider a platoon of N follower vehicles and a leader vehicle labeled 0 traveling in a single lane and exchanging information via V2V communication. For each follower $i\in\mathcal{V}=\{1,\dots,N\}$, the longitudinal dynamics are given by Rajaman (2006):

$$\begin{cases} \dot{p}_i = v_i, \\ \dot{v}_i = a_i, \\ \dot{a}_i = -\dfrac{1}{\tau_i}a_i+\dfrac{1}{\tau_i}\bar{u}_i+f(v_i,a_i), \end{cases} \quad i\in\mathcal{V}, \tag{5}$$

where p_i, v_i and a_i denote the absolute position, velocity and acceleration of vehicle i, respectively. The control input (throttle) is represented

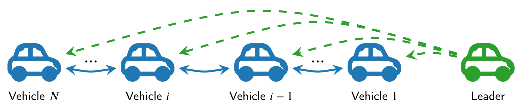

Fig. 1. A typical communication topology in platoon.

by $\bar{u}_i$. τ_i is the engine time constant for vehicle i. The unknown nonlinear function $f(v_i, a_i)$ captures the aerodynamic drag and rolling resistance.

To simplify the controller design, we employ feedback linearization by defining the control law as $\bar{u}_i = u_i - u_{ff}$. Here u_{ff} is a feedforward component to compensate for the nonlinearities. Consequently, the nonlinear dynamics (5) can be transformed into the following state-space representation:

$$\begin{cases} \dot{x}_i = A_\tau x_i + B_\tau u_i + D_i \omega_i, \\ y_i = \begin{bmatrix} p_i - p_{i-1} \\ v_i \end{bmatrix}, \end{cases} \quad i \in \mathcal{V}, \tag{6}$$

where $x_i = \begin{bmatrix} p_i & v_i & a_i \end{bmatrix}^\top \in \mathbb{R}^{n_0}$ is the state vector with $n_0 = 3$. $y_i \in \mathbb{R}^{m_0}$, where $m_0 = 2$, is the locally measurable output, which includes the velocity and the relative position with respect to the preceding vehicle as measured by onboard sensors such as LiDAR. $\omega_i \in \mathbb{R}^{q_i}$ represents the external disturbance acting on vehicle i. $D_i \in \mathbb{R}^{n_0 \times q_i}$ is the known disturbance distribution matrix. A_τ and B_τ are known system matrices for all vehicles, defined as follows:

$$A_\tau = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{\tau} \end{bmatrix}, \; B_\tau = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\tau} \end{bmatrix}, \tag{7}$$

where τ is the nominal engine time constant. In this model, we assume a uniform nominal constant τ for all vehicles to simplify the design process. Any discrepancy between the actual engine constant τ_i and the nominal value τ is treated as model uncertainty and absorbed into the disturbance term $D_i \omega_i$. Therefore, heterogeneous platoons can also be accommodated.

The leader vehicle dynamics are given by:

$$\begin{cases} \dot{x}_0 = A_\tau x_0, \\ y_0 = \begin{bmatrix} p_0 \\ v_0 \end{bmatrix}, \end{cases} \tag{8}$$

where $x_0 = \begin{bmatrix} p_0 & v_0 & a_0 \end{bmatrix}^\top \in \mathbb{R}^{n_0}$ is the leader's state vector. In this paper, we assume that the leader vehicle moves at a constant velocity, i.e., $a_0 = 0$.

As the leader serves as the global reference for the platoon, its state information is crucial for coordination. Therefore, we assume that each follower can directly obtain the leader's state information via V2V communication. With the exception of the leader, the communication among the followers is assumed to be a strongly connected directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Here, $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of nodes representing the following vehicles in the platoon. A typical communication topology is illustrated in Fig. 1.

The main goal is to design the controller u_i to ensure the stability of the platoon, which can be described as follows:

$$\begin{cases} \lim_{t\to\infty} \left(p_i - p_0 + d_{i,0}\right) = 0 \\ \lim_{t\to\infty} \left(v_i - v_0\right) = 0, \\ \lim_{t\to\infty} \left(a_i - a_0\right) = 0, \end{cases} \quad i \in \mathcal{V}, \tag{9}$$

where $d_{i,0}$ is the desired distance with respect to the leader, determined by the chosen spacing policy.

For each follower i, the classical control input u_i is given by Petrillo et al. (2021):

$$u_i = K_i \sum_{j=0}^{N} a_{ij} \left(x_i - x_j + \begin{bmatrix} d_{i,j} & 0 & 0 \end{bmatrix}^\top \right), \tag{10}$$

where a_{ij} is the (i,j)th entry of the adjacency matrix $\mathcal{A}$, and $K_i \in \mathbb{R}^{1 \times n_0}$ is the feedback gain matrix to be designed. $d_{i,j}$ is the desired spacing between vehicles i and j.

Several methods exist to design the controller gain in (10) for platooning (Coppola et al., 2022; Petrillo et al., 2021; Zhao et al., 2021). Most of these methods assume that the desired spacing $d_{i,j}$ is available to each vehicle in the platoon. However, the availability of the desired spacing depends on the chosen spacing policy. Under a constant-spacing (CS) policy, the desired spacing is a fixed constant and is easy to be obtained. By contrast, under the CTH policy in (11), the desired spacing varies with vehicle velocity. Therefore, the velocities of all preceding vehicles must be known.

$$d_{i,j} = (i - j)d + h \sum_{k=j+1}^{i} v_k, \tag{11}$$

where d is the minimum distance and the positive constant h represents the constant headway time. In summary, even with a known spacing policy, computing the control input in (10) is often impractical.

To address this issue, a natural approach is to design an observer that estimates transformed states containing the desired spacing information for the platoon. We then compute the control input from these estimated states. To present the controller based on the estimated states, we first rewrite the dynamics of each vehicle in the platoon.

Under the CTH spacing policy (11), we define the state vector ε_i for each vehicle as follows:

$$\varepsilon_i = \begin{bmatrix} p_i - p_0 + d_{i,0} \\ v_i - v_0 \\ a_i - a_0 \end{bmatrix} \in \mathbb{R}^{n_0}, \quad i \in \mathcal{V}, \tag{12}$$

the dynamics of each vehicle in the platoon can be described as:

$$\dot{\varepsilon}_i = A_{h\tau} \varepsilon_i + A_h \sum_{j=0}^{i-1} \varepsilon_j + B_\tau u_i + D_i \omega_i, \tag{13a}$$

$$u_i = K_i \sum_{j=0}^{N} a_{ij} \left(\varepsilon_i - \varepsilon_j\right), \tag{13b}$$

$$y_i = C_p \varepsilon_{i-1} + C_f \varepsilon_i + C_v v_0 + C_d d, \tag{13c}$$

where the matrices are defined as follows:

$$A_{h\tau} = \begin{bmatrix} 0 & 1 & h \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{\tau} \end{bmatrix}, \; A_h = \begin{bmatrix} 0 & 0 & h \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \; C_f = \begin{bmatrix} 1 & -h & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

$$C_p = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \; C_v = \begin{bmatrix} -h \\ 1 \end{bmatrix}, \; C_d = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

By defining the collective state vector of the platoon as $\varepsilon = \begin{bmatrix} \varepsilon_1^\top & \varepsilon_2^\top & \cdots & \varepsilon_N^\top \end{bmatrix}^\top \in \mathbb{R}^n$, where $n = Nn_0$, collective disturbance $\omega = \begin{bmatrix} \omega_1^\top & \cdots & \omega_N^\top \end{bmatrix}^\top$, and the collective controller input $u = \begin{bmatrix} u_1 & \cdots & u_N \end{bmatrix}^\top$, the overall dynamics of the platoon can be expressed as:

$$\begin{cases} \dot{\varepsilon} = A_\varepsilon \varepsilon + B_\varepsilon u + D\omega, \\ y_i = C_i \varepsilon + C_v v_0 + C_d d, \end{cases} \quad i \in \mathcal{V}, \tag{14}$$

where the matrices are defined as follows:

$$\begin{aligned} & A_\varepsilon = \mathbb{I}_N \otimes A_{h\tau} + H \otimes A_h, \; B_\varepsilon = \mathbb{I}_N \otimes B_\tau, \\ & D = \operatorname{diag}\left(D_1, \; D_2, \; \dots, \; D_N\right), \\ & C_i = \begin{cases} \begin{bmatrix} C_f & 0 & \cdots & 0 \end{bmatrix}, & i = 1, \\ \begin{bmatrix} 0 & \cdots & C_p & C_f & \cdots & 0 \end{bmatrix}, & i \neq 1, \end{cases} \end{aligned} \tag{15}$$

where H is a lower triangular matrix with all entries below and on the main diagonal equal to 1.

Before discussing the observer design, it is essential to analyze the system's observability. For any individual vehicle i, the pair (A_ε, C_i) is typically unobservable, meaning the collective state of the entire platoon cannot be reconstructed solely from local measurements. However, by defining the collective output as $y = [y_1^\top, \ldots, y_N^\top]^\top$, the system is said to be jointly observable if the pair (A_ε, C) is observable, where C represents the augmented output matrix. Leveraging this joint observability, a distributed observer can be developed to estimate the collective state ε by combining local measurements with information shared via communication network (Han et al., 2019). A typical distributed observer is given by:

$$\begin{aligned}\dot{\hat{\varepsilon}}_i &= A_\varepsilon \hat{\varepsilon}_i + B_\varepsilon u + L_i(y_i - \hat{y}_i) - r_i \gamma P_i^{-1} \sum_{j=1}^{N} a_{ij}(\hat{\varepsilon}_i - \hat{\varepsilon}_j), \\ \hat{y}_i &= C_i \hat{\varepsilon}_i + C_v v_0 + C_d d,\end{aligned} \tag{16}$$

where $\hat{\varepsilon}_i \in \mathbb{R}^n$ is the estimated state of the entire platoon computed by the ith distributed observer. $\hat{y}_i$ is the estimated output of vehicle i. $L_i \in \mathbb{R}^{n\times m_0}$ and P_i are observer gain matrices. r_i is a positive scalar from Lemma 1, $\gamma > 0$ is a constant gain, a_{ij} is the adjacency matrix according to the communication topology.

Unlike a centralized observer, the proposed distributed observer incorporates a consensus term (the final term in the equation), which leverages shared information from neighbors to reconstruct states that are locally unobservable.

Previous works on distributed observer design require the full control input u (Han et al., 2019; Mitra & Sundaram, 2018; Wang & Morse, 2018). However, in practical scenarios, each vehicle's control input is private and not shared with other vehicles. Although Yang et al. (2022) proposed a distributed observer that handles unknown control inputs, the platoon system here does not satisfy the required rank conditions.

Therefore, our main goal is to design a distributed observer-controller for each vehicle that removes the need for unknown desired spacing information in the controller and full control inputs in the observer, while ensuring string stability and convergence of the distributed observer. We formalize this via an $\mathcal{H}_\infty$ performance bound as follows:

$$\sqrt{\varepsilon^\top \Omega_\varepsilon \varepsilon + e^\top e} \le \mu \|\omega\|_2. \tag{17}$$

where $\mu > 0$ is the disturbance attenuation level. The collective estimation error is defined as $e = \begin{bmatrix} e_1^\top & \cdots & e_N^\top \end{bmatrix}^\top \in \mathbb{R}^{nN}$, with $e_i = \hat{\varepsilon}_i - \varepsilon \in \mathbb{R}^n$. $\Omega_\varepsilon = \mathrm{diag}\left(v_1, v_2, \ldots, v_N\right) \otimes \mathbb{I}_{n_0}$ is a weighting matrix for the string stability, where $0 < v_i < v_j$ for all $i < j$.

3. Distributed observer–controller co–design

In this section, the distributed observer-controller co-design is presented to achieve the platoon objective (17). First, we introduce the controller structure utilizing the distributed observer, and then propose an observer that leverages estimated control signals. Then, we derive the closed-loop platoon dynamics under the distributed observer-controller. Finally, we provide sufficient conditions for the co-design that guarantee the objective (17).

3.1. Closed-loop dynamics

Replacing the true state in (13b) with its estimate from the distributed observer, the control input for vehicle i is given by:

$$u_i(\hat{\varepsilon}_i) = \sum_{j=0}^{i-1} K_{ij}\left(F_i \hat{\varepsilon}_i - F_j \hat{\varepsilon}_i\right), \tag{18}$$

where $F_j \in \mathbb{R}^{n_0\times n}$ extracts the state of vehicle j from the estimated state of the entire platoon and is defined as:

$$F_j = \begin{bmatrix} 0 & \cdots & \underbrace{\mathbb{I}_{n_0}}_{j\text{-th}} & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{n_0\times n}. \tag{19}$$

As for the distributed observer, although the full control input u is unavailable, the estimated collective state $\hat{\varepsilon}_i$ can be used to reconstruct the control input for vehicle i. Thus, we replace the real control input u in (16) with the estimated control signal $\hat{u}(\hat{\varepsilon}_i)$ and propose the distributed observer as follows:

$$\dot{\hat{\varepsilon}}_i = A_\varepsilon \hat{\varepsilon}_i + B_\varepsilon \hat{u}(\hat{\varepsilon}_i) + L_i(y_i - \hat{y}_i) - r_i \gamma P_i^{-1} \sum_{j=1}^{N} a_{ij}(\hat{\varepsilon}_i - \hat{\varepsilon}_j), \tag{20}$$

where the estimated control signal $\hat{u}(\hat{\varepsilon}_i)$ is

$$\begin{aligned}\hat{u}(\hat{\varepsilon}_i) &= \begin{bmatrix} \hat{u}_1(\hat{\varepsilon}_i) & \cdots & \hat{u}_k(\hat{\varepsilon}_i) & \cdots & \hat{u}_N(\hat{\varepsilon}_i) \end{bmatrix}^\top, \\ \hat{u}_k(\hat{\varepsilon}_i) &= \sum_{j=0}^{k-1} K_{kj}\left(F_k \hat{\varepsilon}_i - F_j \hat{\varepsilon}_i\right).\end{aligned} \tag{21}$$

By substituting the control input (18) into the vehicle dynamics (13a), each vehicle's dynamics can be described as:

$$\begin{aligned}\dot{\varepsilon}_i &= A_{h\tau}\varepsilon_i + A_h \sum_{j=0}^{i-1} \varepsilon_j \\ &\quad + B_\tau \sum_{j=0}^{i-1} K_{ij}(F_i e_i - F_j e_i + \varepsilon_i - \varepsilon_j) + D_i \omega_i.\end{aligned} \tag{22}$$

Similar to (14), the dynamics of the platoon can be described as:

$$\begin{cases} \dot{\varepsilon} = A_\varepsilon \varepsilon + B_\varepsilon u(\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_N) + D\omega, \\ y_i = C_i \varepsilon + C_v v_0 + C_d d, \end{cases} \quad i \in \mathcal{V}, \tag{23}$$

where $u(\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_N)$ is the collective control input of the platoon, defined as $u(\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_N) = \begin{bmatrix} u_1(\hat{\varepsilon}_1) & \cdots & u_N(\hat{\varepsilon}_N) \end{bmatrix}^\top$.

The closed-loop dynamics of the platoon are as follows:

$$\dot{\varepsilon} = A_\varepsilon \varepsilon + B_\varepsilon K \varepsilon + B_\varepsilon K F e + \omega, \tag{24}$$

where the matrices K and F are defined as:

$$\begin{aligned} K &= \begin{bmatrix} K_1^\top & K_2^\top & \cdots & K_k^\top & \cdots & K_N^\top \end{bmatrix}^\top \in \mathbb{R}^{N\times n}, \\ K_k &= \begin{bmatrix} -K_{k1} & \cdots & -K_{k,k-1} & \sum_{j=0}^{k-1} K_{kj} & 0 & \cdots \end{bmatrix} \in \mathbb{R}^{1\times n}, \\ F &= \begin{bmatrix} \bar{F}_1 & \bar{F}_2 & \cdots & \bar{F}_k & \cdots & \bar{F}_N \end{bmatrix} \in \mathbb{R}^{n\times nN}, \\ \bar{F}_k &= \begin{bmatrix} F_k^\top - F_0^\top & F_k^\top - F_1^\top & \cdots & F_k^\top - F_{k-1}^\top & 0 \end{bmatrix}^\top \in \mathbb{R}^{n\times n}. \end{aligned}$$

Before deriving the error dynamics of the distributed observer, we first analyze the term $\Delta u = \hat{u}(\hat{\varepsilon}_i) - u(\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_N)$. Here, Δu is the difference between the control signals reconstructed from the ith observer's estimate $\hat{\varepsilon}_i$ and the true distributed control signals obtained by applying the controller to the local estimates of all vehicles $\hat{\varepsilon}_1, \ldots, \hat{\varepsilon}_N$. For the kth term of Δu, we have

$$\begin{aligned}\Delta u_k &= \hat{u}_k(\hat{\varepsilon}_i) - u_k(\hat{\varepsilon}_k) \\ &= K_k \bar{F}_k e_i - K_k \bar{F}_k e_k.\end{aligned} \tag{25}$$

Therefore, the collective term Δu is given by:

$$\Delta u = \mathcal{K}\mathcal{F} e_i - K F e, \tag{26}$$

where the matrices $\mathcal{K}$ and $\mathcal{F}$ are defined as:

$$\begin{aligned}\mathcal{K} &= \mathrm{diag}\left(K_1, K_2, \ldots, K_N\right) \in \mathbb{R}^{N\times nN}, \\ \mathcal{F} &= \begin{bmatrix} \bar{F}_1^\top & \bar{F}_2^\top & \cdots & \bar{F}_N^\top \end{bmatrix}^\top \in \mathbb{R}^{nN\times n}.\end{aligned}$$

Then, the error dynamics of the distributed observer can be derived as:

$$\begin{aligned}\dot{e}_i &= A_\varepsilon e_i - L_i C_i e_i + B_\varepsilon \mathcal{K}\mathcal{F} e_i - B_\varepsilon K F e \\ &\quad - r_i \gamma P_i^{-1} \sum_{j=1}^{N} a_{ij}(e_i - e_j) - D\omega.\end{aligned} \tag{27}$$

Furthermore, the collective error dynamics of the platoon can be expressed as:

$$\begin{aligned}\dot{e} &= \mathrm{diag}\left(A_\varepsilon - L_i C_i\right) e - \gamma P^{-1}\left(R\mathcal{L} \otimes \mathbb{I}_n\right) e \\ &\quad + \mathrm{diag}_N\left(B_\varepsilon \mathcal{K}\mathcal{F}\right) e - \mathrm{diag}_N\left(B_\varepsilon K\right) \bar{\mathcal{F}} e - \left(\mathbf{1}_N \otimes D\right)\omega.\end{aligned} \tag{28}$$

where $\bar{\mathcal{F}} = \mathbf{1}_N \otimes F$.

The closed-loop dynamics of the platoon under the distributed observer-controller are given by (24) and (28). Compared with the classical observer-based closed loop, the key difference is the presence of the additional terms $B_\epsilon KFe$ in (24) and $\operatorname{diag}_N(B_\epsilon\mathcal{K}\mathcal{F})e - \operatorname{diag}_N(B_\epsilon K)\bar{\mathcal{F}}e$ in (28), which arise from the mismatch between the true and estimated control inputs. To handle these additional terms, we propose the following lemma, built on Lemma 2, for use in the subsequent analysis.

Lemma 3. *For given matrices $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^n$, $K \in \mathbb{R}^{1\times n}$, $F \in \mathbb{R}^{n\times m}$ and $A_e \in \mathbb{R}^{m\times m}$, and symmetric positive definite matrices $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$, if there exist $\iota_1 > 0$ and $\iota_2 > 0$ such that the following inequality holds:*

$$\begin{bmatrix} \operatorname{Sym}(AQ+BKQ) & 0 \\ * & \bar{M}_d \end{bmatrix} < 0, \tag{29}$$

where

$$\bar{M}_d = \begin{bmatrix} \operatorname{Sym}(PA_e) & 0 & F^\top & P & 0 \\ * & -\frac{1}{\iota_1}Q & 0 & 0 & (BKQ)^\top \\ * & * & -\iota_1 Q & 0 & 0 \\ * & * & * & -\frac{1}{\iota_2}\mathbb{I}_m & 0 \\ * & * & * & * & -\iota_2\mathbb{I}_n \end{bmatrix} < 0,$$

then the following inequality is satisfied:

$$\begin{bmatrix} \operatorname{Sym}(AQ+BKQ) & 0 \\ * & \operatorname{Sym}(PA_e + PBKF) \end{bmatrix} < 0. \tag{30}$$

Proof. Since $\bar{M}_d < 0$, applying Lemma 2, where $X = \begin{bmatrix} P^\top & 0 & 0\end{bmatrix}^\top$, $Y = \begin{bmatrix} 0 & BKQ & 0\end{bmatrix}$, and $S = \mathbb{I}$, we obtain

$$\begin{bmatrix} \operatorname{Sym}(PA_e) & PBKQ & F^\top \\ * & -\frac{1}{\iota_1}Q & 0 \\ * & * & -\iota_1 Q \end{bmatrix} < 0. \tag{31}$$

Applying Lemma 2 again, with $X = PBK$, $Y = F$, and $S = Q$, we obtain $\operatorname{Sym}(PA_e + PBKF) < 0$. This completes the proof. □

3.2. Design and convergence analysis

In this subsection, we analyze the stability of the closed-loop platoon system with the distributed observer-controller. The objective is to design the controller gain K_{ij} and the observer gain L_i at the same time to guarantee the $\mathcal{H}_\infty$ performance in (17).

The Lyapunov function candidate is chosen as:

$$V = \varepsilon^\top Q^{-1}\varepsilon + e^\top P e, \tag{32}$$

where the positive definite matrices Q and P are

$$\begin{aligned} Q^{-1} &= \operatorname{diag}\left(Q_1^{-1}, Q_2^{-1}, \dots, Q_N^{-1}\right) \in \mathbb{R}^{n\times n},\ Q_i \in \mathbb{R}^{n_0\times n_0} \\ P &= \operatorname{diag}\left(P_1, P_2, \dots, P_N\right) \in \mathbb{R}^{nN\times nN},\ P_i \in \mathbb{R}^{n\times n}. \end{aligned}$$

Define $W = \dot{V} + \varepsilon^\top\Omega_\varepsilon\varepsilon + e^\top e - \mu^2\|\omega\|_2^2$, where $\dot{V}$ is the time derivative of the Lyapunov function. According to Zemouche and Boutayeb (2009), if $W < 0$, then the $\mathcal{H}_\infty$ performance (17) is satisfied.

To simplify the subsequent analysis, we define the following matrices:

$$\begin{aligned} X_i &= P_i L_i, \\ \Lambda_i &= \operatorname{Sym}\left(P_i A_\varepsilon - X_i C_i\right) \in \mathbb{R}^{n\times n}, \\ \Lambda_p &= \begin{bmatrix} \Lambda_1 + \mathbb{I}_n & \Lambda_2 + \mathbb{I}_n & \cdots & \Lambda_N + \mathbb{I}_n \end{bmatrix} \in \mathbb{R}^{n\times nN}, \\ \Lambda &= \operatorname{diag}\left(\Lambda_1, \Lambda_2, \dots, \Lambda_N\right) \in \mathbb{R}^{nN\times nN}, \\ \Gamma_p &= \begin{bmatrix} D^\top P_1 & D^\top P_2 & \cdots & D^\top P_N \end{bmatrix} \in \mathbb{R}^{q\times nN}, \\ \bar{M}_\varepsilon &= \operatorname{Sym}\left(Q^{-1}A_\varepsilon + Q^{-1}B_\varepsilon K\right). \end{aligned} \tag{33}$$

Calculating W along the trajectory of the closed-loop system, we have:

$$W = W_\varepsilon + W_e - \mu^2\omega^\top\omega, \tag{34}$$

where W_ε and W_e are

$$\begin{aligned} W_\varepsilon =& \varepsilon^\top\left(\bar{M}_\varepsilon + \Omega_\varepsilon\right)\varepsilon + 2\varepsilon^\top Q^{-1}D\omega \\ W_e =& + e^\top\left(\Lambda + \mathbb{I}_{nN}\right)e - 2\gamma e^\top\left(\hat{\mathcal{L}}\otimes\mathbb{I}_n\right)e \\ &+ e^\top\operatorname{Sym}\left(P\left(\operatorname{diag}\left(B_\varepsilon\mathcal{K}\mathcal{F}\right) - \operatorname{diag}\left(B_\varepsilon K\right)\bar{\mathcal{F}}\right)\right)e \\ &- 2\varepsilon^\top Q^{-1}B_\varepsilon KFe - 2e^\top P\left(\mathbf{1}_N\otimes D\right)\omega. \end{aligned}$$

Similar to the results in Yang et al. (2022), the collective error space $e \in \mathbb{E}$ is decomposed into two orthogonal subspaces. One subspace, denoted as $\mathbb{E}_c$, is the kernel of $\hat{\mathcal{L}}\otimes\mathbb{I}_n$. The other subspace is the orthogonal complement of $\mathbb{E}_c$, denoted as $\mathbb{E}_r$, such that $\mathbb{E}_r\oplus\mathbb{E}_c = \mathbb{E}$. Therefore, we denote $e_c \in \mathbb{E}_c$ and $e_r \in \mathbb{E}_r$. From Lemma 1, the kernel of $\hat{\mathcal{L}}\otimes\mathbb{I}_n$ has the form $\mathbf{1}_N\otimes\kappa$, which means $e_c = \mathbf{1}_N\otimes\kappa$. Therefore, W_e is rewritten as:

$$\begin{aligned} W_e =& -2\varepsilon^\top Q^{-1}B_\varepsilon KFe_r - 2\varepsilon^\top Q^{-1}B_\varepsilon KF(\mathbf{1}_N\otimes\kappa) \\ &+ e_r^\top\left(\Lambda + \mathbb{I}_{nN}\right)e_r + (\mathbf{1}_N\otimes\kappa)^\top\left(\Lambda + \mathbb{I}_{nN}\right)(\mathbf{1}_N\otimes\kappa) \\ &+ 2e_r^\top\left(\Lambda + \mathbb{I}_{nN}\right)(\mathbf{1}_N\otimes\kappa) - 2\gamma e_r^\top\left(\hat{\mathcal{L}}\otimes\mathbb{I}_n\right)e_r \\ &- 2e_r^\top P\left(\mathbf{1}_N\otimes D\right)\omega - 2(\mathbf{1}_N\otimes\kappa)^\top P\left(\mathbf{1}_N\otimes D\right)\omega \\ &+ e_r^\top\left(\operatorname{Sym}\left(P\left(\operatorname{diag}\left(B_\varepsilon\mathcal{K}\mathcal{F}\right) - \operatorname{diag}\left(B_\varepsilon K\right)\bar{\mathcal{F}}\right)\right)\right)e_r, \end{aligned} \tag{35}$$

where the following equations are used:

$$\begin{aligned} &\left(\hat{\mathcal{L}}\otimes\mathbb{I}_n\right)e_c = 0, \\ &\left(\operatorname{diag}\left(B_\varepsilon\mathcal{K}\mathcal{F}\right) - \operatorname{diag}\left(B_\varepsilon K\right)\bar{\mathcal{F}}\right)(\mathbf{1}_N\otimes\kappa) = 0. \end{aligned} \tag{36}$$

Note that,

$$\begin{aligned} (\mathbf{1}_N\otimes\kappa)^\top\left(\Lambda + \mathbb{I}_{nN}\right)(\mathbf{1}_N\otimes\kappa) =& \kappa^\top\left(\sum\Lambda_i + N\mathbb{I}_n\right)\kappa, \\ e_r^\top\left(\Lambda + \mathbb{I}_{nN}\right)(\mathbf{1}_N\otimes\kappa) =& e_r^\top\Lambda_p^\top\kappa. \end{aligned} \tag{37}$$

Similarly, we have

$$\begin{aligned} -2\varepsilon^\top Q^{-1}B_\varepsilon KF(\mathbf{1}_N\otimes\kappa) =& -2\varepsilon^\top B_\varepsilon\mathcal{K}\mathcal{F}\kappa, \\ -2(\mathbf{1}_N\otimes\kappa)^\top P(\mathbf{1}_N\otimes D)\omega =& -2\kappa^\top\sum_{i=0}^{N}P_i D\omega. \end{aligned} \tag{38}$$

Then applying $-e_r^\top(\hat{\mathcal{L}}\otimes\mathbb{I}_n)e_r \le -\lambda_2(\hat{\mathcal{L}})e_r^\top e_r$, where $\lambda_2(\hat{\mathcal{L}})$ is the second smallest eigenvalue of $\hat{\mathcal{L}}$, the following inequality is obtained,

$$\begin{aligned} W \le& W_\varepsilon - 2\varepsilon^\top Q^{-1}B_\varepsilon KFe_r - 2\varepsilon^\top Q^{-1}B_\varepsilon\mathcal{K}\mathcal{F}\kappa \\ &+ e_r^\top M_d e_r + \kappa^\top M_c\kappa + 2e_r^\top\Lambda_p^\top\kappa - \mu^2\omega^\top\omega \\ &- 2e_r^\top\Gamma_p^\top\omega - 2\kappa^\top\sum_{i=0}^{N}P_i D\omega \\ &+ e_r^\top\operatorname{Sym}\left(P\left(\operatorname{diag}\left(B_\varepsilon\mathcal{K}\mathcal{F}\right) - \operatorname{diag}\left(B_\varepsilon K\right)\bar{\mathcal{F}}\right)\right)e_r \\ =& \begin{bmatrix}\varepsilon^\top & e_r^\top & \kappa^\top & \omega^\top\end{bmatrix} M_1 \begin{bmatrix}\varepsilon^\top & e_r^\top & \kappa^\top & \omega^\top\end{bmatrix}^\top, \end{aligned} \tag{39}$$

where

$$\begin{aligned} M_1 =& \begin{bmatrix} \bar{M}_\varepsilon + \Omega_\varepsilon & -Q^{-1}B_\varepsilon KF & -Q^{-1}B_\varepsilon\mathcal{K}\mathcal{F} & Q^{-1}D \\ * & M_r & \Lambda_p^\top & -\Gamma_p^\top \\ * & * & M_c & -\sum P_i D \\ * & * & * & -\mu^2\mathbb{I}_p \end{bmatrix}, \\ M_r =& M_d + \operatorname{Sym}\left(P\left(\operatorname{diag}\left(B_\varepsilon\mathcal{K}\mathcal{F}\right) - \operatorname{diag}\left(B_\varepsilon K\right)\bar{\mathcal{F}}\right)\right), \\ M_d =& \Lambda + \mathbb{I}_{nN} - 2\gamma\lambda_2(\hat{\mathcal{L}})\mathbb{I}_{nN},\ M_c = \sum\Lambda_i + N\mathbb{I}_n. \end{aligned}$$

Using pre-and-post-multiplication M_1 by $T = \operatorname{diag}(Q, \mathbb{I}, \mathbb{I}, \mathbb{I})$, we have

$$M_2 = TMT = \begin{bmatrix} M_\varepsilon + Q\Omega_\varepsilon Q & -B_\varepsilon KF & -B_\varepsilon\mathcal{K}\mathcal{F} & D \\ * & M_r & \Lambda_p^\top & -\Gamma_p^\top \\ * & * & M_c & -\sum P_i D \\ * & * & * & -\mu^2\mathbb{I}_p \end{bmatrix}, \tag{40}$$

where $M_\varepsilon = \operatorname{Sym}\left(A_\varepsilon Q - B_\varepsilon KQ\right)$.

If we show that $M_2 < 0$, then $W < 0$, which implies that the $\mathcal{H}_\infty$ performance (17) is satisfied. However, it is difficult to solve for the controller and observer gains from $M_2 < 0$ directly since M_2 contains many non-convex terms, such as $\mathrm{Sym}\big(P\big(\mathrm{diag}\,(B_\varepsilon\mathcal{K}\mathcal{F}) - \mathrm{diag}\,(B_\varepsilon K)\bar{\mathcal{F}}\big)\big)$. To overcome this issue, the following theorem is proposed by applying Lemma 3.

Theorem 1. *Consider the platoon system (23) together with the distributed observer-controller (18) and (20). If there exist positive definite matrices $Q_i \in \mathbb{R}^{n_0\times n_0}$, $P_i \in \mathbb{R}^{n\times n}$, and matrices $X_i \in \mathbb{R}^{n\times m_0}$ and $Y_{ij} \in \mathbb{R}^{1\times n_0}$ for any $i \in \mathcal{V}$, $j \in \{1,\dots,i-1\}$ such that the following convex optimization problem is feasible:*

min (μ) *subject to*

$$\begin{bmatrix} \mathbb{M}_\varepsilon & 0 & \mathbb{D} & \mathbb{Y}_1 & 0 \\ * & \mathbb{M}_d & \mathbb{P} & 0 & \mathbb{F}_1 \\ * & * & \mathbb{M}_c & 0 & \mathbb{F}_2 \\ * & * & * & -\frac{1}{\alpha}\underset{N+1}{\mathrm{diag}}(Q) & 0 \\ * & * & * & * & -\alpha\underset{N+1}{\mathrm{diag}}(Q) \end{bmatrix} < 0, \tag{41}$$

where α, ι_1 and $\iota_2 > 0$ are the given scalars, and the matrices are defined as:

$$\begin{aligned}
&\mathbb{M}_\varepsilon = \begin{bmatrix} M_\varepsilon & Q \\ * & -\Omega_\varepsilon^{-1} \end{bmatrix},\ \mathbb{D} = \begin{bmatrix} 0 & D \\ 0 & 0 \end{bmatrix},\ \mathbb{Y}_1 = \begin{bmatrix} B_\varepsilon Y & B_\varepsilon \mathcal{Y} \\ 0 & 0 \end{bmatrix},\\
&\mathbb{M}_c = \begin{bmatrix} M_c & -\sum P_i D \\ * & -\mu^2 \mathbb{I}_p \end{bmatrix},\ \mathbb{P} = \begin{bmatrix} \Lambda_p^\top & -\Gamma_p^\top \\ 0 & 0 \end{bmatrix},\\
&\mathbb{F}_1 = \begin{bmatrix} F^\top & 0 \\ 0 & 0 \end{bmatrix},\ \mathbb{F}_2 = \begin{bmatrix} 0 & \mathcal{F}^\top \\ 0 & 0 \end{bmatrix},\\
&Y_k = \begin{bmatrix} -Y_{k1} & -Y_{k2} & \cdots & -Y_{k,k-1} & Y_{k,k} & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{1\times n},\\
&Y = \begin{bmatrix} Y_1^\top & Y_2^\top & \vdots & Y_k^\top & \vdots & Y_N^\top \end{bmatrix}^\top \in \mathbb{R}^{N\times n},\\
&\mathcal{Y} = \mathrm{diag}\,(Y_1, Y_2, \dots, Y_N) \in \mathbb{R}^{N\times nN},\\
&\mathbb{M}_d = \begin{bmatrix} M_d & 0 & \mathbb{F}_3 & P & 0 \\ * & -\frac{1}{\iota_1}\underset{N^2+N}{\mathrm{diag}}\,(Q) & 0 & 0 & \mathbb{Y}_2^\top \\ * & * & -\iota_1\underset{N^2+N}{\mathrm{diag}}\,(Q) & 0 & 0 \\ * & * & * & -\frac{1}{\iota_2}\mathbb{I}_{nN} & 0 \\ * & * & * & * & -\iota_2\mathbb{I}_{nN} \end{bmatrix}\\
&\mathbb{F}_3 = \begin{bmatrix} \underset{N}{\mathrm{diag}}(\mathcal{F}^\top) & -(1_N\otimes F)^\top \end{bmatrix},\\
&\mathbb{Y}_2 = \begin{bmatrix} \underset{N}{\mathrm{diag}}(B_\varepsilon\mathcal{Y}) & \mathrm{diag}\,(B_\varepsilon Y) \end{bmatrix},
\end{aligned} \tag{42}$$

then the $\mathcal{H}_\infty$ performance (17) is satisfied. The distributed observer gains are $L_i = P_i^{-1}X_i$. The controller gains are given by

$$\begin{aligned}
&K_{ij} = Y_{ij}Q_j^{-1}, \quad \forall i \in \mathcal{V}, j \in \{1,\ \dots,\ i-1\},\\
&K_{i0} = Y_{ii}Q_i^{-1} - \sum_{k=1}^{i-1} K_{ik}, \quad \forall i \in \mathcal{V}.
\end{aligned} \tag{43}$$

Proof. According to Lemma 3, if the LMI (41) holds, then the following matrix inequality holds:

$$\mathcal{M} = \begin{bmatrix} \mathbb{M}_\varepsilon & 0 & \mathbb{D} & \mathbb{Y}_1 & 0 \\ * & M_r & \begin{bmatrix} \Lambda_p^\top & -\Gamma_p^\top \end{bmatrix} & 0 & \begin{bmatrix} F^\top & 0 \end{bmatrix} \\ * & * & \mathbb{M}_c & 0 & \begin{bmatrix} 0 & \mathcal{F}^\top \end{bmatrix} \\ * & * & * & -\frac{1}{\alpha}\underset{N+1}{\mathrm{diag}}(Q) & 0 \\ * & * & * & * & -\alpha\underset{N+1}{\mathrm{diag}}(Q) \end{bmatrix} < 0. \tag{44}$$

Using Young's inequality on M_2, we have

$$\begin{aligned}
M_2 =& M_3 + \bar{Z}_1\bar{Z}_2^\top + \bar{Z}_2\bar{Z}_1^\top\\
\le& M_3 + \alpha\bar{Z}_1(\underset{N+1}{\mathrm{diag}}(Q))\bar{Z}_1^\top + \frac{1}{\alpha}\bar{Z}_2(\underset{N+1}{\mathrm{diag}}(Q))^{-1}\bar{Z}_2^\top,
\end{aligned} \tag{45}$$

where $\alpha_1 > 0$ and

$$\begin{aligned}
&M_3 = \begin{bmatrix} M_\varepsilon + Q\Omega_\varepsilon Q & 0 & 0 & D \\ * & M_r & \Lambda_p^\top & -\Gamma_p^\top \\ * & * & M_c & -\sum P_i D \\ * & * & * & -\mu^2\mathbb{I}_p \end{bmatrix},\\
&\bar{Z}_1 = \begin{bmatrix} B_\varepsilon K & B_\varepsilon\mathcal{K} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix},\ \bar{Z}_2 = \begin{bmatrix} 0 & 0 \\ F^\top & 0 \\ 0 & \mathcal{F}^\top \\ 0 & 0 \end{bmatrix}.
\end{aligned} \tag{46}$$

Then, by applying the Schur lemma to deal with Q^2 in M_3, we have

$$M_2 \le \begin{bmatrix} M_4 & Z_1 & Z_2 \\ * & -\frac{1}{\alpha}\underset{N+1}{\mathrm{diag}}(Q) & 0 \\ * & * & -\alpha\underset{N+1}{\mathrm{diag}}(Q) \end{bmatrix} = \mathcal{M} < 0, \tag{47}$$

where M_4 Z_1 and Z_2 are defined as

$$\begin{aligned}
&M_4 = \begin{bmatrix} \mathbb{M}_\varepsilon & 0 & \mathbb{D} \\ * & M_r & \begin{bmatrix} \Lambda_p^\top & -\Gamma_p^\top \end{bmatrix} \\ * & * & \mathbb{M}_c \end{bmatrix},\\
&Z_1 = \begin{bmatrix} B_\varepsilon Y & B_\varepsilon\mathcal{Y} \\ 0 & 0 \\ 0 & 0 \end{bmatrix},\ Z_2 = \begin{bmatrix} 0 & 0 \\ F^\top & 0 \\ 0 & \mathcal{F}^\top \end{bmatrix}.
\end{aligned} \tag{48}$$

Therefore, $M_2 < 0$ holds, which implies that $W < 0$. The proof is completed. □

Algorithm 1 outlines the procedure for the proposed distributed observer-controller co-design method.

Algorithm 1 The co-design procedure of distributed observer–controller.

Require: The number of vehicles in platoon N, the laplacian matrix $\mathcal{L}$ of the communication $\mathcal{G}$, the minimum distance d, the constant headway time h and the engine time τ.

1: Choose the parameters α, ι_1, ι_2;

2: Get Y_{ij} and X_i by solving the convex optimization problem in the Theorem 1;

3: Compute the observer gain $L_i = P_i^{-1}X_i$;

4: Compute the controller gain K_{ij} according to (43);

3.3. Discussion

In this subsection, we discuss the practical implementation of the proposed method, specifically focusing on its reliance on global information and its scalability.

According to Theorem 1, the observer and controller gains are determined through a centralized offline design. This implies that the gains are pre-calculated and fixed before deployment. During the design stage, the distributed observer relies on the second smallest eigenvalue λ_2 of the Laplacian matrix, rather than the details of individual communication links. Since the platoon's communication topology is typically pre-defined, this eigenvalue is generally accessible to the designer. If necessary, existing adaptive techniques can be employed to estimate this value online, further reducing the need for prior global knowledge (Kim et al., 2020).

In distributed estimation theory, a strongly connected graph is a necessary condition to ensure that the state information of all agents is reachable by every local observer. While estimating the states of the entire platoon increases communication bandwidth and computational requirements as the platoon scales, this "comprehensive view" offers significant advantages for high-level objectives, such as fault tolerance and resilience against cyber-attacks.

Once the observer and controller gains are determined, they are deployed locally on each vehicle. During real-time operation, each vehicle processes local measurements and neighbor data within a consistent computational framework, maintaining a uniform observer structure across the platoon.

As analyzed in Section 2, the primary motivation of this work is to eliminate the explicit dependency of distributed controllers on desired-spacing information when ensuring string stability. In practical applications, the desired spacing is often treated as high-level decision data, typically provided by Roadside Infrastructure (RSI) or the leader vehicle. However, such methods are inherently centralized. If the RSI fails, the platoon may lose access to critical desired-spacing information, which can seriously compromise safety. For this reason, vehicles should possess the capability to determine the desired spacing autonomously, which is essential for reliable and safe platoon operation. Consequently, we adopt a distributed observer-based control architecture. While this approach provides the necessary autonomy and resilience, it entails a trade-off in the form of increased communication bandwidth requirements for the distributed observers.

4. Simulation on QLabs with QCar2

To validate the effectiveness of the proposed method, we conduct a simulation using the Self-Driving Car Studio in QLabs. QLabs is a virtual environment that simulates the dynamics of the physical QCar2 platform, which is a 1/10th scale autonomous vehicle. The Self-Driving Car Studio provides a realistic simulation environment for testing and validating autonomous driving algorithms, including platooning control strategies.

In the following subsections, we first describe the simulation setup, including the vehicle dynamics, the communication topology, and the implementation of the distributed observer-controller. Then, we present the results of three scenarios:

1. the leader maintaining a constant velocity on a smooth road to test the stability of the platoon;
2. the leader starting and stopping on a smooth road to test the transient performance of the platoon and the string stability;
3. the platoon moving on the wave road to test the robustness of the platoon.

4.1. Description of the simulation setup

The QCar2 is actuated by drive and steering servo motors. A 6-axis IMU, LiDAR, and wheel encoders are used to measure the QCar2's states. Further details are provided in Quanser (2025). We consider the open-road scenario, which spans more than 50 km of multi-lane roads with long, flat straight sections, which are ideal for implementing platooning control, as shown in Fig. 2.

The vehicle dynamics of the QCar2 platform are inherently nonlinear, as characterized by (5). To bridge the gap between the physical system and the linearized model used for controller synthesis, a feedforward component u_{ff} is designed to compensate for these nonlinearities. Specifically, the steady-state mapping between the throttle input $\bar{u}_i$ and the vehicle velocity v_i was identified through a series of open-loop experiments. By applying various step throttle signals and performing a linear regression on the resulting steady-state data, the feedforward control law is determined as $u_{ff} = 0.156385 v_i + 0.005230$. Furthermore, the engine time constant is set to $\tau_{\text{real}} = 0.16$ s, according to the technical note from Quanser. It should be noted that while u_{ff} provides primary compensation, residual modeling uncertainties and uncompensated nonlinearities are lumped into the exogenous disturbance term ω_i in (6). The leader's trajectory is generated by PID speed controller to follow a reference velocity.

To compute the state estimates with the distributed observer running on each vehicle, the outputs y_i include the distance to the preceding vehicle and the vehicle velocity. These signals are measured by the onboard LiDAR and wheel encoders, respectively, and are corrupted by sensor noise. Inter-vehicle communication follows the prescribed communication topology. It is implemented over hostPort using User Datagram Protocol (UDP) to emulate wireless links among the physical QCar2 platforms. The communication update rate is set to 20 Hz.

We adopt the CTH policy with minimum distance $d = 0.8$ m and headway time $h = 1$ s. The parameters in the Theorem 1 are set to $\alpha = 1$, $\iota_1 = 1$ and $\iota_2 = 1$. To assess the robustness against parameter uncertainty, a nominal engine time constant of $\tau = 0.1$s is adopted during the design process, allowing us to test the platoon's performance under model mismatch. Unless otherwise specified, the communication topology is shown as Fig. 1. By solving the LMI (41) using the YALMIP toolbox in MATLAB, the coupling gain and disturbance attenuation level are $\gamma = 347.64$ and $\mu = 140.9307$. The controller gains K_{ij} are

$$
\begin{aligned}
K_{10} &= \begin{bmatrix} -0.382 & -1.01 & -0.209 \end{bmatrix},\\
K_{20} &= \begin{bmatrix} -0.514 & -0.668 & -0.233 \end{bmatrix},\\
K_{21} &= \begin{bmatrix} -0.0139 & -0.0352 & 0.0649 \end{bmatrix},\\
K_{30} &= \begin{bmatrix} -0.564 & -0.753 & -0.297 \end{bmatrix},\\
K_{31} &= \begin{bmatrix} -0.0196 & -0.0511 & 0.0379 \end{bmatrix},\\
K_{32} &= \begin{bmatrix} -0.0292 & -0.0623 & 0.0453 \end{bmatrix}.
\end{aligned} \tag{49}
$$

The distributed observer gains L_i are shown in (50).

$$
\begin{aligned}
L_1^\top &= \begin{bmatrix} 22.1 & -24.9 & 10.4 & 31.6 & 14.5 & 1.2 & 53.8 & 28.8 & -4.45 \\ 121 & 103 & 31.5 & 29.1 & 1.68 & 2.73 & 6.54 & -5.54 & 2.51 \end{bmatrix},\\
L_2^\top &= \begin{bmatrix} 0.512 & 9.75 & -2.22 & 35.9 & -32.2 & 6.02 & 106 & 55.8 & -4.32 \\ -129 & -61.9 & -33.9 & 256 & 250 & 64.1 & 124 & 57.6 & -1.68 \end{bmatrix},\\
L_3^\top &= \begin{bmatrix} 5.59 & 15.4 & -0.647 & 18.4 & 11.8 & -8.93 & 54.3 & -37.5 & 4.85 \\ -30 & -42.3 & -36.4 & -104 & -6.5 & -37.6 & 436 & 456 & 95.1 \end{bmatrix}.
\end{aligned} \tag{50}
$$

Unless otherwise specified, the communication topology, as illustrated in Fig. 1, is employed for the following three scenarios. The states of the distributed observers are initialized to zero for all vehicles. At $t = 0$, the entire platoon is at a standstill; the leader (Vehicle 0) is positioned at $p_0(0) = 7.5$ m, and each subsequent follower is placed 1.5 m behind its predecessor. Upon the establishment of communication links, the distributed observer-based controllers are simultaneously activated across all vehicles to achieve the desired platoon coordination.

Remark 1. Due to hardware constraints, real-vehicle platooning experiments are currently infeasible. However, to ensure simulation fidelity, key components-including sensor readouts, feedforward controller design, leader speed tracking, and communication network setup-have been rigorously validated using both physical QCar2 platforms and their QLabs digital twins. Consequently, the simulation results reliably reflect actual physical vehicle performance.

4.2. Leader with constant velocity on the smooth road

To validate the stability of the platoon under the proposed distributed observer-controller, we conduct a baseline scenario where the leader vehicle maintains a constant of 0.45 m/s velocity on a smooth road.

Due to space constraints, Fig. 3 only illustrates the estimation performance of the distributed observer for the first following vehicle. Overall, the estimated states exhibit high consistency in tracking the actual vehicle trajectories. During the first 5 sec, a large estimation error is observed, which is primarily attributed to the discrepancy between the initial static positions and the prescribed desired formation during the communication initialization phase. This mismatch induces a significant initial deviation between the observer states and the physical system. However, once the platoon maneuvers commence, the estimation errors rapidly converge to the actual values within 10 sec. Consequently, the platoon successfully achieves and maintains the desired steady-state inter-vehicle spacing, as shown in Fig. 4.

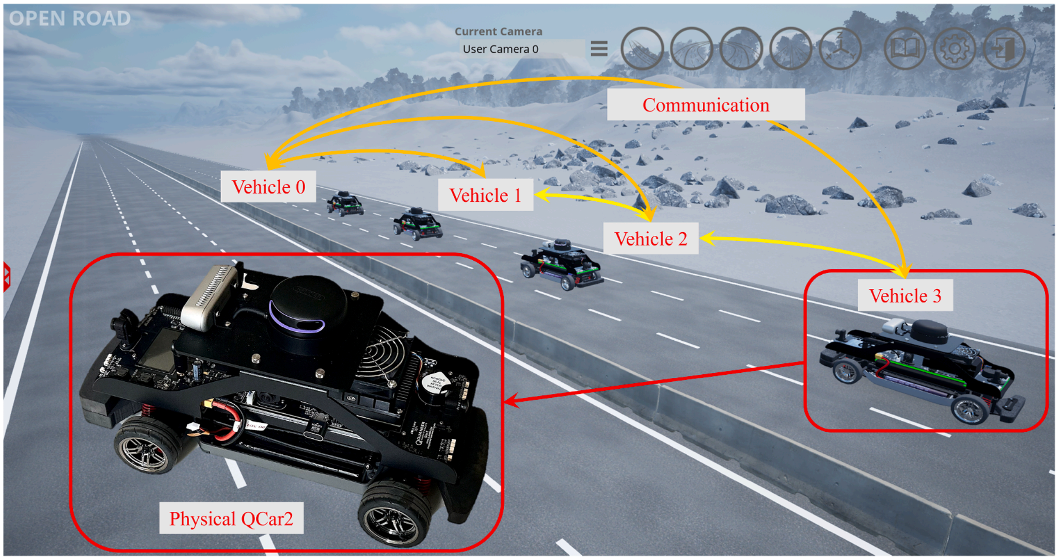

Fig. 2. The QLabs environment.

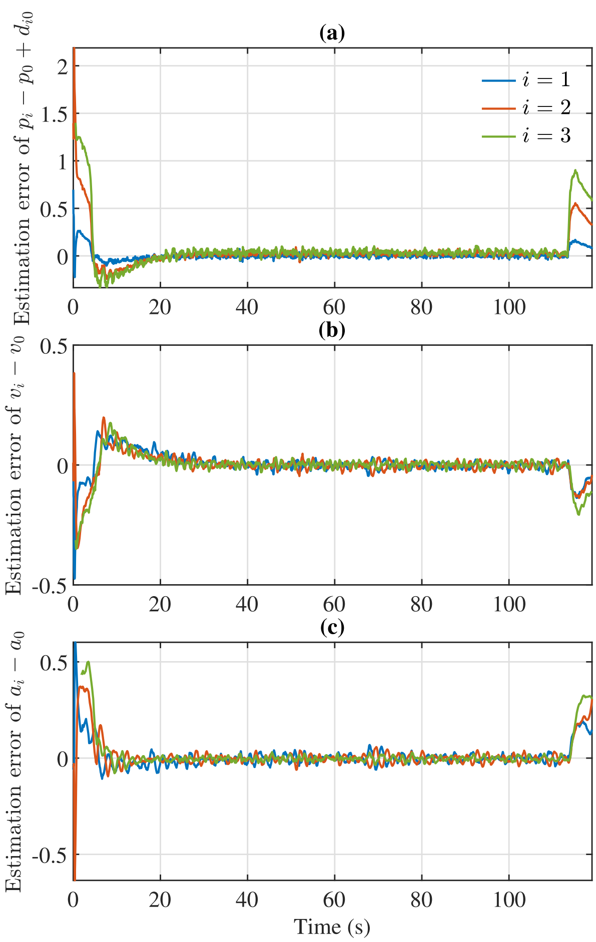

Fig. 3. The estimation results of the distributed observer 1: (a) estimation error of $p_i - p_0 + d_{i0}$; (b) estimation error of $v_i - v_0$; (c) estimation error of $a_i - a_0$.

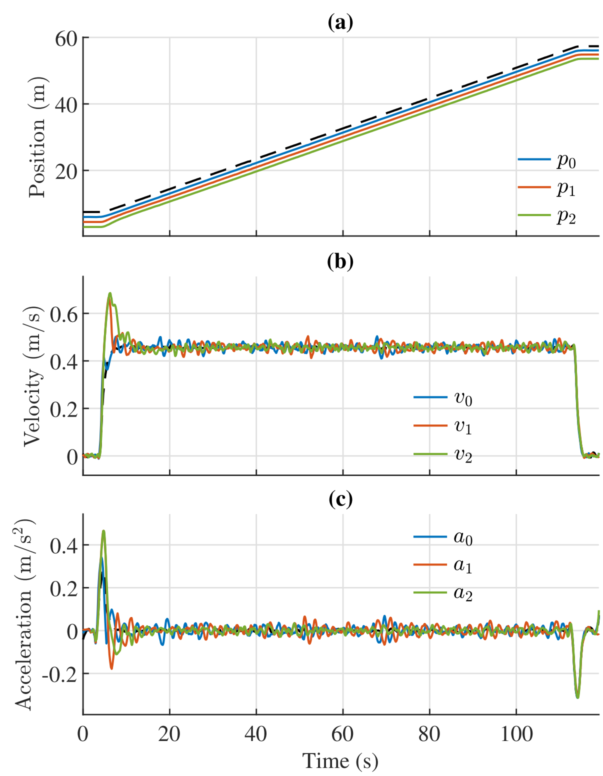

Fig. 4. The trajectory of the platoon: (a) position; (b) velocity; (c) acceleration.

4.3. Platoon on the wave road

To further validate the robustness of the proposed method, we construct a section of wave road for the platoon. When the platoon traverses this section, the vertical disturbance from the uneven surface induces acceleration disturbances in each vehicle. Therefore, the performance of the platoon on the wave road reflects the disturbance attenuation capability of the proposed method.

To build this scenario, the wave road is constructed using 34 thin cuboids, each measuring 0.5 m × 0.8 m × 0.01 m, placed consecutively to simulate a wavy surface. The height of each cuboid is determined by a cosine function, resulting in a total road length of 16.5 m, a wavelength of 0.4125 m, and an amplitude of 0.1 m. The white cuboids in Fig. 5 represent the wave road, where the upper sub-figure depicts the wavy road profile at an original scale.

Fig. 5. The wave road.

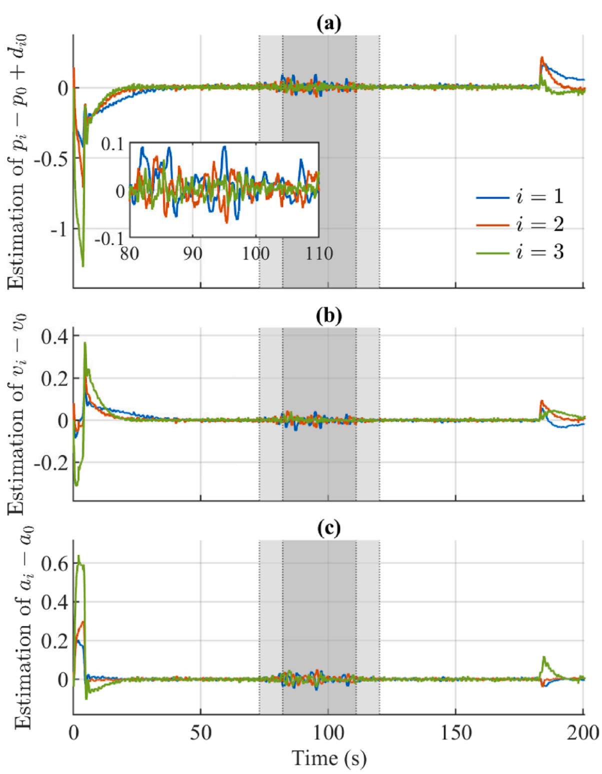

Fig. 6. The estimation results of the distributed observer 3 on the wave road: (a) estimation error of $p_i - p_0 + d_{i0}$; (b) estimation error of $v_i - v_0$; (c) estimation error of $a_i - a_0$.

The estimation results of the distributed observer 3 and the trajectory of the platoon are shown in Figs. 6 and 7, where the occurrence of the wave road is highlighted in the gray area.

Due to the external disturbances introduced by the wave road, the observer estimates exhibit noticeable fluctuations. During the wave-road traversal, the estimation error of the position-related state remains within ±0.1 m, the velocity error within ±0.05 m/s, and the acceleration error within ±0.05 m/s^2. It is worth noting that at the beginning and the end of the wave-road section (when the platoon has not fully entered or fully left the wave road), the estimates converge quickly to the true values once the leader leaves the wave road, even though some follower vehicles are still on it. This behavior is mainly because the distributed observer uses information from neighbor vehicles that are less affected by the wave road, which helps the distributed observer reach consensus and correct the estimates more quickly.

As shown in the true vehicle states in Fig. 7, the wave road inevitably induces oscillations in the vehicle dynamics. Facilitated by the

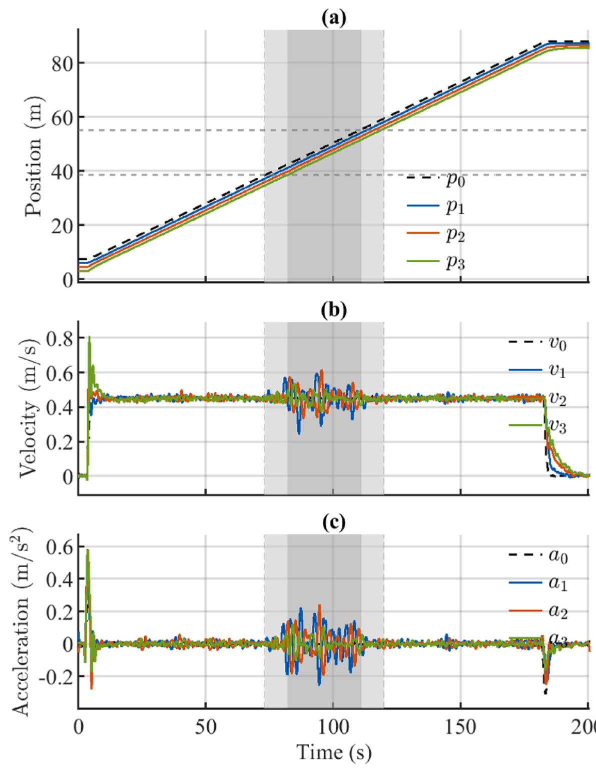

Fig. 7. The true vehicle states on the wave road: (a) position; (b) velocity; (c) acceleration.

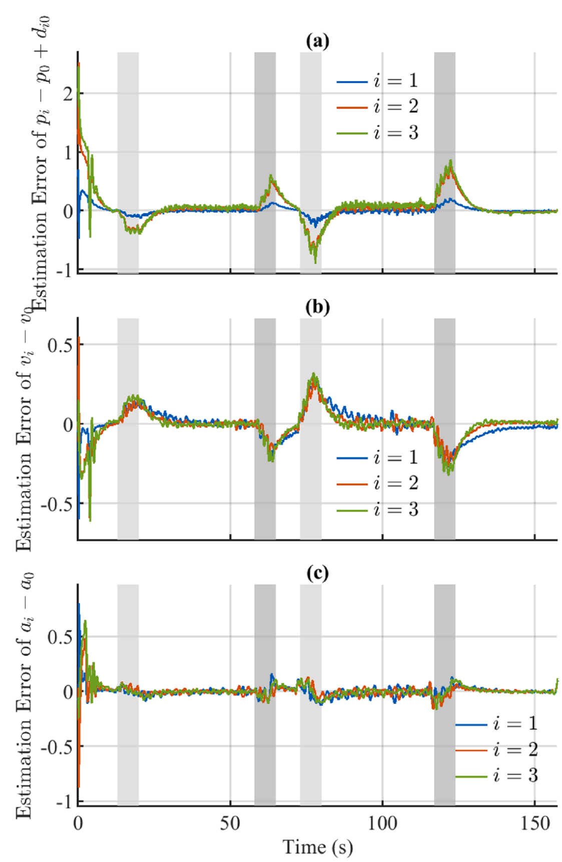

Fig. 8. The estimation error of the first distributed observer under the speed change scenario: (a) estimation error of $p_i - p_0 + d_{i0}$; (b) estimation error of $v_i - v_0$; (c) estimation error of $a_i - a_0$.

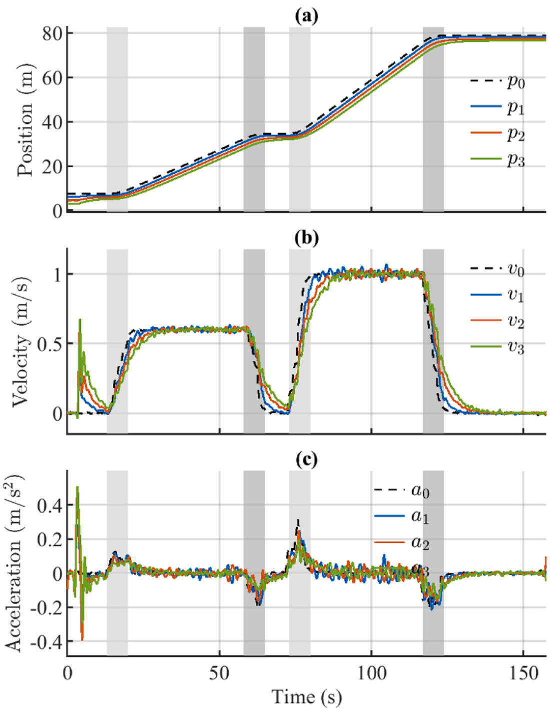

Fig. 9. The true vehicle states under the speed change scenario: (a) position; (b) velocity; (c) acceleration.

distributed observers, the platoon rapidly restores equilibrium upon entering and exiting the undulating section. Moreover, as depicted in the velocity and acceleration profiles in Fig. 7 (b) and (c), the oscillation amplitudes progressively attenuate from the leading to the trailing vehicle. This downstream attenuation confirms that the platoon preserves string stability despite the road-induced disturbances.

4.4. Leader starts and stops on the smooth road

In real driving, the leader does not always maintain a constant speed. When the leader brakes or accelerates quickly, the followers must respond quickly and maintain string stability to avoid collisions. To evaluate the proposed method in this case, we consider two start–stop profiles on a smooth road: a mild profile and an aggressive profile. In the first profile, the leader accelerates at 0.067 m/s^2 to 0.6 m/s, cruises for a period, and then decelerates at −0.057 m/s^2 to 0 m/s. In the second profile, the leader accelerates at 0.13 m/s^2 to 1 m/s, cruises for a period, and then decelerates at −0.15 m/s^2 to 0 m/s.

In this scenario, the observer estimation results and the true vehicle states are shown in Figs. 8 and 9, respectively. The gray area marks the leader start–stop phase. It can be seen that, due to the assumption that the leader acceleration is zero, the distributed observer has larger estimation errors during the start–stop phase. A larger leader acceleration leads to a larger error. Even with this mismatch, the controller still maintains weak string stability of the platoon. As shown in Fig. 9 (c), during the leader start–stop phase, the acceleration oscillations decrease from the first follower to the last follower, which indicates that string stability is preserved. These results support the effectiveness and robustness of the proposed method.

5. Conclusion

This paper proposed a distributed observer-controller co-design methodology to ensure the string stability of vehicle platoons subject to external disturbances and incomplete spacing information. A distributed observer reconstructs each vehicle's full state using local measurements and neighbor communication without requiring other vehicles' private control inputs. The observer and controller are co-designed via a convex optimization with LMI constraints that guarantee exponential convergence of the estimation error and an $\mathcal{H}_\infty$ disturbance attenuation bound for the platoon. The effectiveness of this methodology was validated through high-fidelity simulations in QLabs across three scenarios. The results demonstrate that the platoon successfully maintains string stability and avoids collisions, even when faced with external disturbances such as leader start-stop maneuvers and uneven road profiles. Future research will focus on extending this design to account for communication delays, packet drops, and fault injection, ultimately leading to validation on physical hardware.

CRediT authorship contribution statement

Shengya Meng: Writing – review & editing, Writing – original draft, Validation, Software, Methodology; **Ali Zemouche:** Writing – review & editing, Methodology, Funding acquisition; **Marouane Alma:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shengya MENG reports financial support was provided by China Scholarship Council. Ali ZEMOUCHE reports financial support was provided by ANR agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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